Tracing Authorial Identity Through Iterative Paraphrasing

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Abstract

Repeated machine paraphrasing can preserve a text's broad meaning while altering linguistic signals of its origin. We examine the Ship of Theseus Paraphrased Corpus across three questions: which features change first (RQ1), how original-source attribution degrades (RQ2), and whether paraphrasers leave detectable fingerprints (RQ3). A lexical and semantic baseline covers 3,718 complete Human-origin Yelp/XSum chains; expanded experiments cover 132,659 complete chains across seven datasets, Human and six LLM source labels, and seven paraphraser settings. POS, dependency, and stylometric changes accumulate from T0 to T3. A T0-trained Human–LLM classifier loses most of its accuracy at T1 but remains above chance at T3. At T1, four-way paraphraser identification achieves 63.82% random-forest accuracy. We distinguish degraded source-category recognition from loss of individual authorship.

Processing code, notebooks, reproducibility instructions, and result tables are in https://github.com/xiaoyancai/Ship-of-Theseus-Paraphrased-Corpus-CS6120-Team4/tree/project-update-2. Large processed feature files require the linked external download or regeneration from raw data.

1 Introduction

Large language models can repeatedly rewrite a passage without necessarily changing its central meaning. The Ship of Theseus analogy asks whether a text retains its identity as vocabulary, syntax, and stylistic cues are successively replaced. Beyond semantic similarity, this matters for AI-text forensics and source attribution. We study original documents (T0) and three sequential rewrites (T1–T3) from the Ship of Theseus Paraphrased Corpus [6], using ChatGPT, PaLM, Dipper, and Pegasus with multiple rewriting strengths. Our research questions are: **RQ1**, which lexical, stylometric, and syntactic features change first; **RQ2**, how many rounds weaken recognition of the original *Human versus LLM* source category; and **RQ3**, whether a model can identify the paraphraser from its output. These tasks should not be conflated. A text can be lexically dissimilar to its source yet retain contextual content; source-category detection may fail even if some stylistic traces remain; and the rewriting system can imprint a new pattern without restoring the original author. Studying these distinct outcomes is more informative than trying to define a single numerical point where the text "becomes somebody else's." Here, "source identity" means an original-source category, not an individual writer, and "paraphraser" means the transformation system rather than the source generator.

2 Background and Related Work

1. Authorship Attribution for Neural Text Generation. Uchendu et al. [1] studied attribution of machine-generated text. They tested human–machine classification and generator identification using eight neural generators. Higher-quality outputs confused classifiers more often, making attribution harder.

2. Threads of Subtlety: Detecting Machine-Generated Texts Through Discourse Motifs. Kim et al. [2] examined discourse differences between human and machine writing. They extracted motifs from hierarchical document representations for classification. Discourse features improved detection, including on paraphrased text.

3. A Multifaceted Framework to Evaluate Evasion, Content Preservation, and Misattribution in Authorship Obfuscation Techniques. Altakrori et al. [3] evaluated authorship obfuscation. They tested stronger attribution models across topics and measured evasion, content preservation, and misattribution. Simple back-translation remained competitive across these objectives.

4. Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval Is an Effective Defense. Krishna et al. [4] studied detection after paraphrasing. They developed DIPPER and evaluated rewritten text against AI-text detectors. Paraphrasing substantially weakened detection while largely preserving meaning.

5. BERTScore: Evaluating Text Generation with BERT. Zhang et al. [5] investigated semantic evaluation beyond word overlap. They proposed BERTScore using contextual token embeddings and compared it with human judgments. It better captured similarity between differently worded texts than traditional overlap metrics.

6. A Ship of Theseus: Curious Cases of Paraphrasing in LLM-Generated Texts. Tripto et al. [6] examined authorship signals under repeated paraphrasing. They built a corpus linking original texts to successive machine rewrites and tested attribution. Classification performance declined across paraphrasing rounds.

7. On the Evaluation Metrics for Paraphrase Generation. Shen et al. [7] assessed automatic paraphrase metrics. They compared lexical and neural scores against human judgments and proposed ParaScore. BLEU and ROUGE did not consistently reflect human preferences.

8. Paraphrase Generation and Evaluation on Colloquial-Style Sentences. Sjöblom et al. [8] studied multilingual colloquial paraphrases. They generated text with RNN and Transformer models and compared BLEU and BERTScore with human ratings. BERTScore correlated better at sentence level.

9. DetectGPT: Zero-Shot Machine-Generated Text Detection Using Probability Curvature. Mitchell et al. [9] investigated zero-shot AI-text detection. They proposed DetectGPT, comparing text likelihood with perturbed alternatives. Detection weakened as more text was revised.

10. Adversarial Authorship Attribution for Deobfuscation. Zhai et al. [10] studied recovering attribution from obfuscated writing. They adversarially trained classifiers on transformed examples. This reduced obfuscation effectiveness, suggesting some authorial signals can be recovered.

11. Topic or Style? Exploring the Most Useful Features for Authorship Attribution. Sari et al. [11] compared topical and stylistic authorship cues. They evaluated feature representations across four datasets. Feature effectiveness varied by dataset, and their model improved performance on two datasets.

12. Idiosyncratic but not Arbitrary: Learning Idiolects in Online Registers Reveals Distinctive yet Consistent Individual Styles. Zhu and Jurgens [12] examined individual writing styles across online registers. They trained neural authorship representations using cross-author comparisons and perturbations. The representations captured distinctive, consistent stylistic differences.

13. TinyStyler: Efficient Few-Shot Text Style Transfer with Authorship Embeddings. Horvitz et al. [13] studied few-shot authorship style transfer. They combined a small language model with pretrained authorship embeddings. TinyStyler outperformed stronger baselines, including GPT-4, with fewer parameters.

14. JAMDEC: Unsupervised Authorship Obfuscation using Constrained Decoding over Small Language Models. Fisher et al. [14] investigated authorship obfuscation while preserving meaning. They developed JAMDEC using constrained decoding with small models. It outperformed earlier small-model methods and competed with larger systems.

15. A Large-Scale Computational Study of Content Preservation Measures for Text Style Transfer and Paraphrase Generation. Babakov et al. [15] evaluated content-preservation metrics. They compared 57 measures across 19 datasets and introduced Mutual Implication Score. Cross-encoders generally outperformed traditional metrics.

3 Dataset and Processing

The corpus contains seven domains: CMV, ELI5, SciGen, TLDR, WP, XSum, and Yelp. Original labels are Human or one of six LLM source families (OpenAI, PaLM, LLAMA, BigScience, Tsinghua, and Eleuther-AI). Each chain comprises an original T0 and sequential paraphrases T1–T3. Four paraphraser families appear in seven configurations: ChatGPT, PaLM, Dipper (standard, low, high), and Pegasus (slight, full). The released PaLM source name is used consistently in the analysis.

We normalize `version_name` into stages, align on dataset, document key, source and paraphraser, and keep only complete nonempty four-stage chains. The expanded dataset has **132,659 chains**, or **530,636 stage records**. Feature extraction parses 411,463 unique text strings after exact deduplication, then joins features back to stage records. The 23-dimensional feature vector is complete for POS/dependency measures; 48 stage records (0.009%) have one or more missing word-based features. Model pipelines fit median imputation on training data only. The `dataset::key` group identifier keeps all source versions and paraphraser settings for a document key within one split, although unrelated keys could still represent related text.

For a distinct baseline, we use Human-origin Yelp and XSum with four canonical paraphrasers. This yields 3,718 complete chains (1,956 Yelp; 1,762 XSum); 3,276 chains from 819 source keys have all four paraphrasers and form the matched-source comparisons. XSum PaLM has only 340 complete chains, a coverage limitation rather than a performance judgment. The baseline is a subset, not additional independent data. The chain-completeness requirement preserves T0–T3 alignment but excludes documents with missing paraphrases, so available texts may not represent all original sources equally. Original T0 copies under multiple paraphrasers are deduplicated before RQ2 classifier training; they remain separate matched-chain references for RQ1. The same distinction matters when comparing stage-level sample counts. In the underlying input files, the Human-source Yelp subset contains 10,781 Human-labeled raw rows from 491 unique keys and XSum contains 10,054 Human-source rows from 476 keys. The XSum PaLM coverage shortfall means it should not be directly compared against models with more complete chains without matching documents. All reported paired lexical comparisons therefore use the 819 keys with outputs from each of the four canonical paraphrasers.

4 Methods

4.1 Features and RQ1 Comparisons

Five rule-based features measure type–token ratio (TTR), average sentence length, population variance of sentence length, Unicode punctuation per 100 word tokens, and average word length. Word tokenization lowercases Unicode letters, retains internal apostrophes, and excludes numeric tokens; sentence boundaries use punctuation and line breaks. The 17 universal POS-tag proportions and a dependency-depth measure use English spaCy `en_core_web_sm`. For each sentence, depth is the maximum number of dependency-head links to its root; a document's depth is the mean of sentence maxima.

We compare each rewritten stage to its own T0. BLEU, ROUGE-1/2/L F1, and BERTScore F1 evaluate lexical overlap and contextual similarity on the Yelp/XSum baseline. The current scoring code defaults to `distilbert-base-uncased` without baseline rescaling; the original saved score CSVs do not document their complete runtime configuration. For each of the five style features, T0-relative retention is

$$R_f(g) = 1 - \frac{|f(T_g) - f(T_0)|}{|f(T_g)| + |f(T_0)|}, \tag{1}$$

with retention set to 1 when both values are zero. We equally average the five values into a descriptive style index; missing inputs are not imputed for this index.

For POS frequencies q_p over 17 tags, grammatical drift is the total-variation distance

$$D_{\mathrm{POS}}(g) = \tfrac{1}{2} \sum_p |q_p(T_g) - q_p(T_0)|. \tag{2}$$

We also calculate signed and absolute changes in dependency depth. Each metric is averaged over aligned chains by stage and paraphraser. Human–LLM structural contrasts first average within dataset–paraphraser cells and then give each of the 49 cells equal weight. For individual tags, the stage-wise mean *signed frequency change* is the T1/T2/T3 cohort mean minus its matched T0 cohort mean; opposite signs can cancel.

4.2 RQ2: Original-Source Attribution

We fit a class-balanced radial-basis SVM ($C = 1$) to the 23 features of *T0 only*, then test that unchanged model at each stage. Distinct `group_id` values are split 80/20, stratified by dataset with seed 42 (2,416 training groups; 604 test groups). The training set contains 15,469 deduplicated T0 originals. Held-out T0 has 3,873 unique originals, whereas each later stage contains 26,598 paraphrase variants. Training-only preprocessing uses median imputation and scaling. We report accuracy, balanced accuracy, macro-F1, and Human-class metrics, with descriptive 95% intervals from 2,000 held-out group bootstrap samples. Crucially, the differing T0 versus later-stage row weighting limits strictly paired comparisons.

For complementary geometry, we aggregate features by dataset, group, source category, and stage, then fit training-group imputation, standardization, and two-component PCA. Human and LLM centroids are averaged equally across the seven datasets, and we measure their distance. A 2,800-record balanced t-SNE visualization is exploratory, not a clustering or identity-loss test.

4.3 RQ3: Paraphraser Fingerprints

At T1 only, we classify ChatGPT, PaLM, standard Dipper, and Pegasus(full) outputs. This excludes Dipper(low/high), Pegasus(slight), and later stages. There are 74,643 variants (59,662 train; 14,981 test) using the same key-disjoint group split. We compare majority prediction, class-balanced logistic regression with standardized features, and a class-balanced random forest with 200 trees and minimum leaf size 3. Only the 23 linguistic features are inputs; source, dataset, group, and paraphraser identifiers are excluded. Hyperparameters are fixed without test-set tuning. Logistic regression uses class weights to reduce the impact of the unequal label counts, while the random forest uses class-balanced weights with its prespecified tree size and leaf constraint. Neither the class counts nor train/test group labels are predictive inputs. This evaluation deliberately tests differences in the *linguistic output* rather than using metadata that could trivially identify the paraphraser.

5 Results

5.1 RQ1: Style, Syntax, and Content Retention

In matched Human-source Yelp/XSum chains, lexical overlap declines at each rewrite, while BERTScore remains comparatively high. On XSum, ChatGPT BLEU decreases from .262 (T1) to .201 (T3), versus BERTScore F1 .892 to .874; on Yelp, BLEU drops from .179 to .127, versus BERTScore .869 to .845. At T3, Dipper yields the lowest BLEU (.084 XSum; .073 Yelp), while PaLM has the highest XSum BLEU (.435). The matched-source T3 values are given in Table 1; for example, the PaLM XSum BLEU of .435 co-occurs with BERTScore .919, whereas Dipper reaches .084 BLEU and .823 BERTScore. Within every dataset–paraphraser cell BERTScore is numerically higher than ROUGE-L, but these metrics have different distributions and should not be compared as if they were calibrated to the same retention scale. Taken together, the results demonstrate substantial surface rewriting without establishing that semantics is uniformly invariant. These numbers are metric- and sample-specific, not a cross-metric estimate of semantic decay or a general model-quality ranking.

Table 1: Matched Human-source Yelp/XSum T3 scores. R-L is ROUGE-L F1; BERT is BERTScore F1. Metrics have different scales.

Dataset	Paraphraser	BLEU	R-L	BERT	Style
XSum	ChatGPT	.201	.447	.874	.882
	PaLM	.435	.652	.919	.908
	Dipper	.084	.245	.823	.882
	Pegasus	.248	.516	.852	.855
Yelp	ChatGPT	.127	.390	.845	.850
	PaLM	.203	.480	.864	.856
	Dipper	.073	.250	.814	.856
	Pegasus	.221	.505	.865	.834

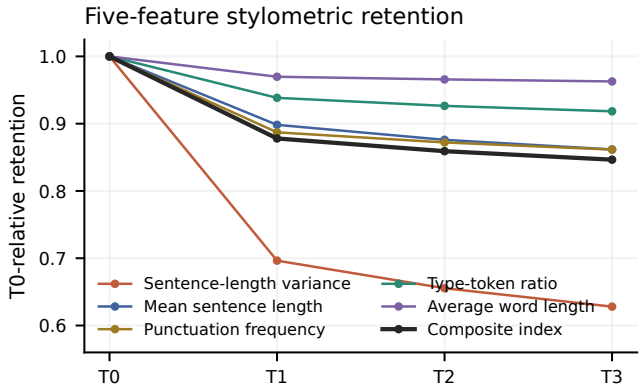


Figure 1: Seven-domain, all-source T0-relative retention of the five initial stylometric features and their equally weighted index.

The new seven-domain stylometric analysis confirms an early shift followed by continued movement (Figure 1). Mean composite style retention falls from 1 at T0 to .878 (T1), .859 (T2), and .846 (T3). Sentence-length variance changes most (T3 retention .628), while average word length is most stable (.963). Signed T3 changes include a 1.594-word decrease in mean sentence length and a .027 increase in TTR. The signed change analysis also reports reduced sentence-length variance (mean difference −70.977 at T3), a .815 increase in punctuation per 100 word tokens, and almost unchanged average word length (−.086). In combination, these results suggest a tendency toward shorter, more regular sentences, though the corpus-level means do not imply that every chain follows this direction. These are differences in measured style, not proof of meaning loss.

POS distribution drift and absolute dependency-depth change both increase from T1 through T3 for every paraphraser configuration (Figure 2). Dipper(high) reaches the greatest T3 values (POS

Table 2: T3 structural drift by paraphraser relative to its T0. Absolute and signed depth report distinct quantities.

Paraphraser	POS drift	$\lvert\Delta d\rvert$	Δd
ChatGPT	.126	1.000	+.349
PaLM	.110	.862	+.102
Dipper	.149	1.025	−.330
Dipper(low)	.096	.867	−.049
Dipper(high)	.281	1.963	−.518
Pegasus(slight)	.072	.565	−.442
Pegasus(full)	.148	1.337	−1.246

.281; absolute depth 1.963); Pegasus(slight) reaches the smallest (.072; .565). Signed depth changes differ even at similar magnitudes: ChatGPT averages +.349 links at T3, but Pegasus(full) averages −1.246. Thus structural change need not be a uniform simplification or expansion.

Figure 3 resolves *which* categories drive average POS changes: determiners (DET) rise by .0111 from T0 to T3, punctuation (PUNCT) by .0080, while proper nouns (PROPN) fall by .0090 and adverbs (ADV) by .0086. DET rises from .0795 to .0906 in absolute frequency, and PROPN falls from .0495 to .0405. Most signed shifts appear by T1; averaging may conceal text-level reversals. In contrast to total-variation POS drift, which is nonnegative on each chain and measures all 17 POS tags, a signed frequency chart shows the *direction* of selected category changes. The two views answer different questions: total variation tells us how much the overall grammatical profile changes, and selected signed changes help explain which categories account for the net mean redistribution. When dataset–paraphraser cells are equally weighted, T3 POS drift is .132 for Human origins versus .140 for LLM origins; absolute depth shifts are .973 versus 1.106. The all-record pooled averages, by contrast, give POS drift .135 (Human) versus .142 (LLM), illustrating how weighting changes a summary. The equal-cell contrasts are more appropriate for a balanced descriptive comparison, although they cannot remove all text-length or source-composition differences. These modest differences do not establish a general origin-specific rule.

5.2 RQ2: Source Attribution and Feature-Space Trajectory

The T0-trained SVM's balanced accuracy falls from .800 at T0 to .669 at T1, .646 at T2, and .631 at T3 (Figure 4; Table 3). Macro-F1 falls .693, .553, .539, and .535. The largest adjacent macro-F1 decline (−.140) is at T1; subsequent declines are smaller. Similarly, balanced accuracy decreases by .130 between T0 and T1 but only .039 from T1 to T3. The first rewrite is therefore the most disruptive stage *for this trained model and evaluation scheme*, not necessarily the point at which all useful author information disappears. At T3, the balanced-accuracy bootstrap interval is [.620,.642], still above .5 chance. **We do not observe a definitive point of no return through T3** for this binary source-category task.

Low Human precision, particularly .232 at T3, reflects difficult recognition under strong LLM-source imbalance. Stage-level subgroup scores also vary (T3 balanced accuracy .573 on TLDR versus .716 on CMV). Since T0 uses deduplicated originals and later stages

Table 3: RQ2: held-out attribution scores (604 document groups at every stage).

Metric	T0	T1	T2	T3
Accuracy	.774	.624	.613	.614
Balanced accuracy	.800	.669	.646	.631
Macro-F1	.693	.553	.539	.535
Human recall	.838	.735	.693	.655
Human precision	.394	.252	.239	.232

Table 4: RQ3: four-class T1 held-out metrics.

Model	Accuracy	Macro-F1
Majority baseline	.259	.103
Logistic regression	.591	.578
Random forest	.638	.623

pool variants, these values represent model transfer to rewritten samples, not a strictly matched change for each writer. The experiment also cannot determine a stage when the *individual writer* becomes unidentifiable: Human/LLM is the only trained label, and the six LLM source families are pooled. Its high Human recall should be read together with low Human precision, rather than as evidence of reliable individual classification.

PCA shows a parallel but narrower result (Figure 5). The first two axes explain 24.45% and 9.91% of feature variation. The dataset-balanced Human–LLM centroid distance shrinks from .275 (T0) to .146 (T1), .142 (T2), and .139 (T3), an overall 49.4% reduction. Centroid movement does *not* prove that all texts cluster together: two axes omit most feature variation and do not measure within-class spread.

5.3 RQ3: Paraphraser Identification

The random forest reaches **63.82% accuracy** and .623 macro-F1 on 14,981 group-held-out T1 rewrites, exceeding logistic regression (59.12%, .578) and a majority-label baseline (25.85%, .103; Table 4). These are preliminary within-corpus results for four canonical classes, not proof that any arbitrary rewriting system can be identified. Because the model receives neither the dataset label nor the source label, its success suggests that measurable grammatical and stylometric cues contain some information about the transformation mechanism. However, performance may also vary with the domains and source categories represented in this training sample.

Pegasus is most recognizable (recall .770), followed by ChatGPT (.729) and Dipper (.668); PaLM is hardest (.353). Of 3,416 true PaLM outputs, 943 are mistaken for ChatGPT and 781 for Dipper (Figure 6). Exploratory standardized logistic coefficients associate ChatGPT with average word length, Dipper with mean sentence length, and Pegasus with punctuation frequency; PaLM has positive punctuation and sentence-length-variance associations. More specifically, the largest positive standardized coefficient per class is average word length for ChatGPT (+1.258), POS punctuation for PaLM (+.546), mean sentence length for Dipper (+.948), and punctuation frequency for Pegasus (+1.076). Because the proportions sum to one and punctuation features overlap with each other,

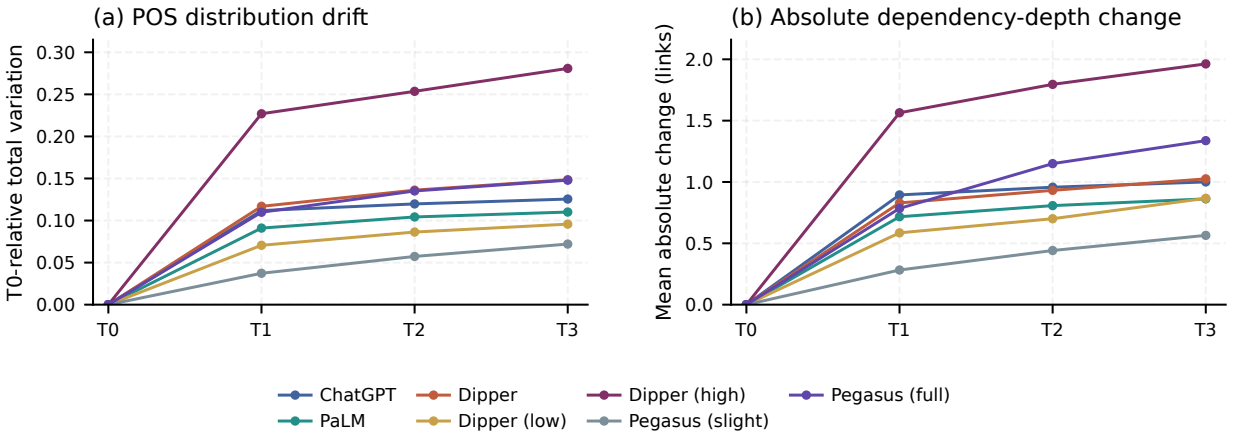


Figure 2: RQ1: mean T0-relative POS total-variation drift and absolute dependency-depth change for all seven rewriting settings. All four stages are shown for complete chains.

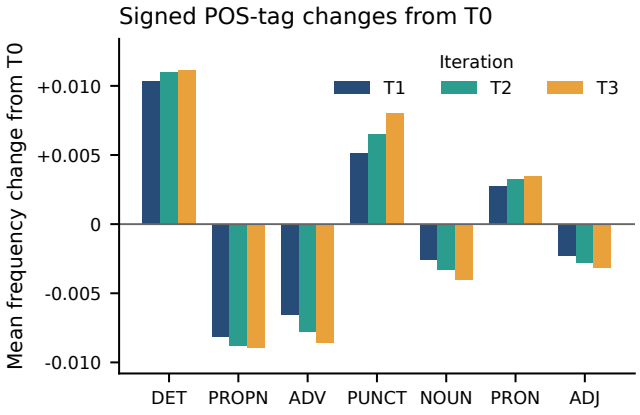


Figure 3: Mean signed POS-tag frequency changes relative to T0 across T1–T3, using both Human- and LLM-origin texts. The seven tags are selected by their absolute mean T3 changes.

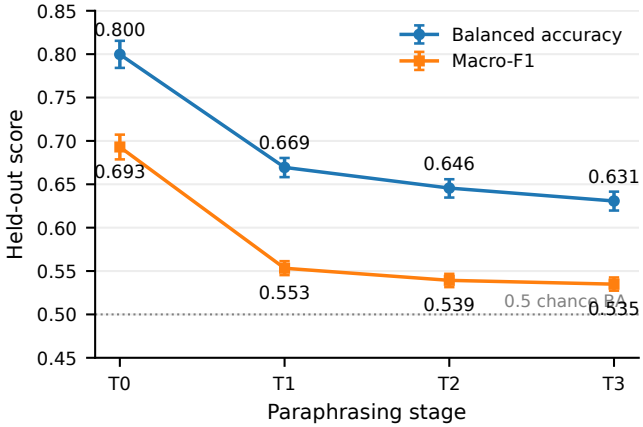


Figure 4: RQ2: held-out T0-trained Human–LLM source-category metrics; error bars are group-bootstrap 95% intervals. The .5 chance reference applies to balanced accuracy only.

coefficients are conditional on a correlated feature set. These *associations* are not causal features or explanations of the random forest. The checked-in RQ3 report includes group-split auditing, per-class precision/recall/F1, and test-set confusion matrices.

6 Discussion

6.1 Origin Cues Versus Rewriting Cues

The results expose two distinct signals. RQ1 finds accumulating lexical and structural drift, especially with stronger paraphrasers; RQ2 finds that the original Human–LLM source category becomes harder to recognize after the first rewrite but remains detectable at T3; RQ3 finds that the rewriting mechanism itself can leave a different recognizable signature. A transformation fingerprint is not evidence of retained individual-author identity. In particular, source attribution and paraphraser identification use different targets and evaluation samples: the former transfers a classifier trained only on T0 across stages, whereas the latter trains and tests on T1 with four paraphraser classes. Stronger RQ3 performance therefore does not contradict weakening RQ2 accuracy. POS changes concern grammatical-category distributions, whereas BLEU/ROUGE reflect word reuse; neither alone demonstrates altered propositional meaning.

6.2 Differences Across Paraphrasers

Intensity is an important source of heterogeneity. Dipper(high) changes POS distribution and dependency depth much more than Dipper(low). Pegasus(full) and Pegasus(slight) also differ substantially. Signed dependency-depth changes suggest that some models lengthen the parsed dependency paths while others shorten them, even when absolute change grows for all models. Thus, pooling different settings into one universal “paraphrasing effect” would

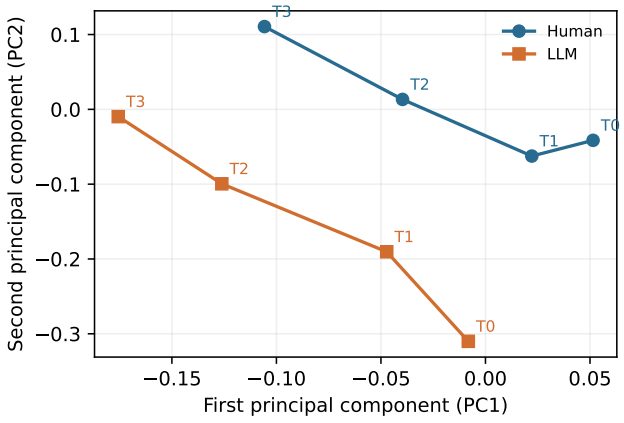


Figure 5: RQ2: Human and LLM category centroids in shared PCA space, balanced across seven datasets; labels mark T0–T3.

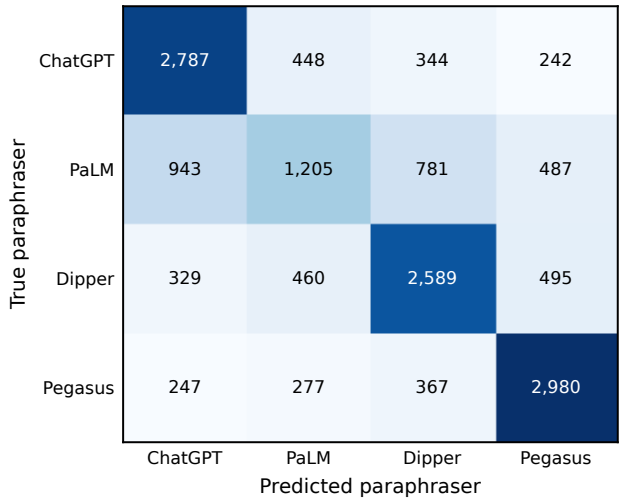


Figure 6: RQ3: random-forest confusion matrix for T1 paraphraser labels; rows are truth and columns predictions.

conceal meaningful directional variation. Likewise, the small source-group differences in RQ1 are not equivalent to an original-source classifier: a mean drift difference need not be predictive of the Human or LLM class at the individual-record level.

7 Limitations

Data and measurement. Complete-chain selection excludes unavailable rewrites and may favor more extensively paraphrased sources. The lexical and semantic baseline covers Human-origin Yelp and XSum, while the expanded structural and classification analyses cover seven domains and Human plus six LLM origin labels; these samples are not interchangeable. Rule-based sentence boundaries can be affected by abbreviations, quotes, and irregular punctuation, and spaCy dependency depth depends on an English parser that may be less accurate on unusual text. BERTScore is a contextual similarity proxy rather than a factual-consistency or entailment check, and the five-feature style index uses arbitrary equal weights. Its numeric scale should not be interpreted as a probability that authorship has survived.

Evaluation and interpretation. RQ2 learns a binary source-category distinction, not individual authors or the six model origins separately. Its source imbalance, the unique-T0-versus-pooled-variant weighting difference, and possible related texts stored under different keys limit causal claims about loss of identity. Group-bootstrap intervals measure uncertainty for the chosen split, but not model-selection or domain-shift uncertainty. PCA summarizes only 34.36% of variation in two axes and does not quantify within-class clustering. RQ3 is a single key-disjoint T1 test with four canonical labels; its feature correlations make standardized coefficients descriptive, not explanations of what the random forest learned. External-domain tests, later paraphrase rounds, and paired resampling are needed before broader forensic claims can be made.

8 Conclusion and Next Steps

Across three paraphrasing rounds, linguistic measurements change progressively, while original-source attribution falls most sharply after T1. Yet the remaining T3 signal rules out a demonstrated point of complete source-category identity loss; separately, T1 paraphraser classification succeeds substantially above a majority baseline. The next priority is a matched-original RQ2 evaluation with equal stage weights, followed by paired uncertainty analyses of POS and domain effects. We also plan to validate meaning preservation against human judgments or entailment-style checks, quantify within-class PCA dispersion, and evaluate RQ3 at T2/T3 and on held-out domains. These changes would more directly test which information disappears, which survives, and which is introduced by the paraphraser.

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